Even with the FFT, the time required to calculate the Fourier transform is a tremendous bottleneck in image processing. For example, the Fourier transform of a 512×512 image requires several minutes on a personal computer. This is roughly 10,000 times slower than needed for real time image processing, 30 frames per second. This long execution time results from the massive amount of information contained in images. For comparison, there are about the same number of *pixels* in a typical image, as there are *words* in this book. Image processing via the frequency domain will become more popular as computers become faster. This is a twenty-first century technology; watch it emerge!

FFT Convolution

Even though the Fourier transform is slow, it is still the fastest way to convolve an image with a large filter kernel. For example, convolving a 512×512 image with a 50×50 PSF is about 20 times faster using the FFT compared with conventional convolution. Chapter 18 discusses how FFT convolution works for one-dimensional signals. The two-dimensional version is a simple extension.

We will demonstrate FFT convolution with an example, an algorithm to locate a predetermined pattern in an image. Suppose we build a system for inspecting one-dollar bills, such as might be used for printing quality control, counterfeiting detection, or payment verification in a vending machine. As shown in Fig. 24-11, a 100×100 pixel image is acquired of the bill, centered on the portrait of George Washington. The goal is to search this image for a known pattern, in this example, the 29×29 pixel image of the face. The problem is this: given an acquired image and a known pattern, what is the most effective way to locate where (or if) the pattern appears in the image? If you paid attention in Chapter 6, you know that the solution to this problem is *correlation* (a matched filter) and that it can be implemented by using *convolution*.

Before performing the actual convolution, there are two modifications that need to be made to turn the target image into a PSF. These are illustrated in Fig. 24-12. Figure (a) shows the target signal, the pattern we are trying to detect. In (b), the image has been rotated by 180°, the same as being flipped left-for-right and then flipped top-for-bottom. As discussed in Chapter 7, when performing *correlation* by using *convolution*, the target signal needs to be reversed to counteract the reversal that occurs during convolution. We will return to this issue shortly.

The second modification is a trick for improving the effectiveness of the algorithm. Rather than trying to detect the face in the original image, it is more effective to detect the *edges of the face* in the *edges of the original image*. This is because the edges are sharper than the original features, making the correlation have a sharper peak. This step isn't required, but it makes the results significantly better. In the simplest form, a 3×3 edge detection filter is applied to both the original image and the target signal